Nour Khalifa

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EDUCATION

Louisiana State University | B.S. Computer Science, Software Engineering | GPA: 3.9 **Expected May 2027**
Chevron Energy Leaders Scholarship (x3), President's Honor Roll (x2)

CERTIFICATIONS

Cornell ML Foundations (Break Through Tech) | IBM Generative & Agentic AI | watsonx Orchestrate | watsonx Code Assistant | watsonx.governance | watsonx.data ETL/ELT | watsonx.data

WORK EXPERIENCE

IBM Baton Rouge Future Now Center | Data Engineering Intern **May 2026 – July 2026**

- Built ETL pipeline processing 3 data sources (CSV, JSON, SQL) into analytics warehouse with 5 fact/dimension tables.
- Delivered healthcare NLP solution classifying 40+ medical specialties across 4,000+ clinical transcriptions with semantic search.
- Developed business innovation capstone using agentic AI for industrial symbiosis, advancing to top 5 out of 28 teams for executive presentation.

LSU ITS | Student DevOps Analyst **Dec. 2024 – Present**

- Managed IT service requests and created technical documentation to streamline processes and improve user experience for students, faculty, and staff.

Break Through Tech AI, Remote | Machine Learning/AI Fellow **May 2024 – April 2025**

- Selected from 3,000+ applicants for year-long fellowship supporting women and underrepresented groups in AI/ML.
- Led team of four on KPMG AI Studio project developing donor behavior prediction model, achieving 98% accuracy.

PROJECTS

NovaCart ETL Data Pipeline | Python, pandas, SQL

- Implemented Bronze-Silver-Gold medallion architecture with idempotent execution; engineered schema validation detecting drift and duplicates; designed SCD Type 1/2 dimensions for star schema.

Medical Specialty Classification & Semantic Search | Python, scikit-learn, FAISS

- Preprocessed clinical text with TF-IDF and chi-square feature selection; trained Logistic Regression, Naive Bayes, and ensemble models; built semantic search using SentenceTransformers indexed with FAISS.

Workflow Status Change Tracking & Analysis | Python, NetworkX, Docker

- Modeled 523 workflow transitions as directed graph with 13 status nodes; implemented depth-first search for reprocess detection; quantified $1.08M recoverable waste from 14,185 failed visits (34.8% failure rate).
- Engineered CLI orchestration layer; containerized pipeline with Docker; designed XGBoost cancellation risk classifier foundation.

SKILLS

Programming: Python, Java, JavaScript, SQL | **Data & ML:** pandas, NumPy, scikit-learn, NLP, TF-IDF, FAISS, NetworkX | **Data Engineering:** ETL pipelines, star schema, Bronze-Silver-Gold architecture | **Tools:** Docker, Git, Linux

ACTIVITIES AND LEADERSHIP

LSU Women in Computer Science | Chevron Leadership Academy | Break Through Tech AI | Rewriting the Code